

Install Docker on Amazon Linux 2 / Amazon Linux 2023 and Connect to the EC2 Instance via SSH to prepare for Docker installation.

```
ubuntu@ip-172-31-16-106: ~  
C:\Users\Md Rustam\Downloads>ssh -i docker_1662026.pem ubuntu@54.81.253.159  
The authenticity of host '54.81.253.159 (54.81.253.159)' can't be established.  
ED25519 key fingerprint is SHA256:PStMliMTSSu0JlQCfokCWew8uJIykDm/SwUZtA/597w.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '54.81.253.159' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Tue Jun 16 00:00:03 UTC 2026  
  
System load:  0.02      Temperature:    -273.1 C  
Usage of /:   30.6% of 6.61GB  Processes:      119  
Memory usage: 27%      Users logged in: 0  
Swap usage:   0%         IPv4 address for ens5: 172.31.16.106  
  
ubuntu@ip-172-31-16-106:~$ cat /etc/os-release  
PRETTY_NAME="Ubuntu 26.04 LTS"  
NAME="Ubuntu"  
VERSION_ID="26.04"  
VERSION="26.04 LTS (Resolute Raccoon)"  
VERSION_CODENAME=resolute  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=resolute  
LOGO=ubuntu-logo  
ubuntu@ip-172-31-16-106:~$ |
```

Update packages the ubuntu latest packages

```
ubuntu@ip-172-31-16-106:~$ sudo apt update -y; sudo apt upgrade -y  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease  
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease  
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease  
All packages are up to date.  
The following packages were automatically installed and are no longer required:  
  bridge-utils containerd dns-root-data dnsmasq-base pigz ubuntu-fan  
Use 'sudo apt autoremove' to remove them.  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
ubuntu@ip-172-31-16-106:~$ |
```

Created an script for Install prerequisites & add Docker repo

```
ubuntu@ip-172-31-16-106:~$ cat docker_repo.sh  
sudo apt update -y  
sudo apt install -y ca-certificates curl gnupg  
sudo install -m 0755 -d /etc/apt/keyrings  
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \  
  sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg  
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \  
  https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo $VERSION_CODENAME) stable" | \  
  sudo tee /etc/apt/sources.list.d/docker.list  
ubuntu@ip-172-31-16-106:~$ |
```

```
-rw-rw-r-- 1 ubuntu ubuntu 457 Jun 16 00:15 docker_repo.sh
ubuntu@ip-172-31-16-106:~$ sudo sh docker_repo.sh
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
All packages are up to date.
ca-certificates is already the newest version (20260223).
ca-certificates set to manually installed.
curl is already the newest version (8.18.0-1ubuntu2.1).
curl set to manually installed.
gnupg is already the newest version (2.4.8-4ubuntu3).
gnupg set to manually installed.
The following packages were automatically installed and are no longer required:
  bridge-utils containerd dns-root-data dnsmasq-base pigz ubuntu-fan
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

Install Docker Engine

```
ubuntu@ip-172-31-16-106:~$ sudo apt install -y docker-ce docker-ce-cli containerd.io
Package docker-ce is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
```

Verify the installed docker version

```
ubuntu@ip-172-31-16-106:~$ docker --version
Docker version 29.5.3, build d1c06ef
```

Start, enable Docker service and verify the status

```
ubuntu@ip-172-31-16-106:~$ sudo systemctl enable --now docker
ubuntu@ip-172-31-16-106:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-06-16 00:29:03 UTC; 15s ago
 Invocation: 00a465cfc5574f7dada90d8dccbdb21d
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 6695 (dockerd)
     Tasks: 9
  Memory: 96.4M (peak: 96.7M)
     CPU: 523ms
    CGroup: /system.slice/docker.service
            └─6695 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

Add ec2-user to docker group (no sudo needed)

```
ubuntu@ip-172-31-16-106:~$ sudo usermod -aG docker ubuntu; newgrp docker
ubuntu@ip-172-31-16-106:~$ cat /etc/group |grep docker
docker:x:109:ubuntu
```

Verify Installation (Confirm Docker is installed and running).

```
ubuntu@ip-172-31-16-106:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
ubuntu@ip-172-31-16-106:~\$ docker pull ubuntu:24.04				
24.04: Pulling from library/ubuntu				
cb259a83ac3d: Pull complete				
183e6e0268de: Download complete				
Digest: sha256:786a8b558f7be160c6c8c4a54f9a57274f3b4fb1491cf65146521ae77ff1dc54				
Status: Downloaded newer image for ubuntu:24.04				
docker.io/library/ubuntu:24.04				

```
ubuntu@ip-172-31-16-106:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
ubuntu:24.04	786a8b558f7b	119MB	31.7MB	

Run and control containers interactively or in background.

```
ubuntu@ip-172-31-16-106:~$ docker run -it ubuntu:24.04 bash
root@6dc6d4bf9a46:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@6dc6d4bf9a46:/#
```

```
ubuntu@ip-172-31-16-106:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
6dc6d4bf9a46	ubuntu:24.04	"bash"	44 seconds ago	Exited (0) 9 seconds ago		adoring_wozniak

Work with Volumes (Persist data across container restarts using volumes).

```
ubuntu@ip-172-31-16-106:~$ docker volume create vol_01
vol_01
ubuntu@ip-172-31-16-106:~$ docker run -d -v vol_01:/data ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
58b0b5e8885f37d60993756ef74a2553a1c01a7a0249ed3b1d72d58e8274e9cb
ubuntu@ip-172-31-16-106:~$
```

```
ubuntu@ip-172-31-16-106:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
58b0b5e8885f	ubuntu	"/bin/bash"	About a minute ago	Exited (0) 59 seconds ago		infallible_edison
6dc6d4bf9a46	ubuntu:24.04	"bash"	4 minutes ago	Exited (0) 3 minutes ago		adoring_wozniak

Verify the mapped Docker volume and test persistence between the host node and the container

```
ubuntu@ip-172-31-16-106:/var/lib$ docker volume ls
```

DRIVER	VOLUME NAME
local	vol_01

```
ubuntu@ip-172-31-16-106:/var/lib$
```

```

root@ip-172-31-16-106:~# cd /var/lib/docker/volumes/vol_01
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01# ls -l
total 4
drwxr-xr-x 2 root root 4096 Jun 16 00:39 _data
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01# cd _data/
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01/_data# ls -l
total 0
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01/_data# mkdir dir_1; touch file.txt
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01/_data# ls -l
total 4
drwxr-xr-x 2 root root 4096 Jun 16 00:47 dir_1
-rw-r--r-- 1 root root    0 Jun 16 00:47 file.txt
root@ip-172-31-16-106:/var/lib/docker/volumes/vol_01/_data# |

```

Explore Networking (Connect containers using custom networks).

```

ubuntu@ip-172-31-16-106:/var/lib$ docker network create mynet
f2f41ba028b28b31f80f16e57e216ec443f6dd0263142d9a95cc73eae9b76e06
ubuntu@ip-172-31-16-106:/var/lib$ docker run -d --name web --network mynet nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
6376488be516: Pull complete
72c03230f136: Pull complete
d4dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
9e02e6386301a34d0e78bb3a9f9cba9765016f48a0509da68e128e89c64eeea1
ubuntu@ip-172-31-16-106:/var/lib$ |

ubuntu@ip-172-31-16-106:/var/lib$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9e02e6386301	nginx	"/docker-entrypoint..."	34 seconds ago	Up 34 seconds	80/tcp	web